

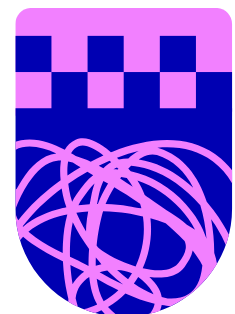
Foundations of AI Regulation on Social Media

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Complex Social &
Computational Systems



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Why think about legal foundations?

Finding Normative Grounding

Bits from yesterday

- We should design aligned swarms. **Aligned to what?**
- What if half of the world wants a different thing? **What thing do we want?**
- More general: **Which capabilities should we aim to design/train "into" (or out of) AI swarms?**

"When aiming to improve societal outcomes, some computer science for social good work lacks a rigorous foundation in politics and social change." [1]

→ Anchoring normative commitments in legal texts provides such a foundation and sets the scope (EU).

[1] Green, Ben. "Good" isn't good enough. Proceedings of the AI for Social Good workshop at NeurIPS Vol. 17, 2019.

Why Social Media?

The harms of social media platforms are well-documented^[1], they...



... foster the spread of extreme opinions and misinformation [2, 3].



... contribute to increasing polarisation and segregation of perceived reality (filter bubbles) [4, 5].



... reduce political knowledge and social trust [6, 7].



... concentrate influence and visibility in a small elite [8].

AI swarms could supercharge each of these problems.

[1] Scharfbillig et al., [Fractured reality](#), *Publications Office of the European Union* (2026).

[2] Müller & Schwarz, "[Fanning the Flames of Hate](#)", *J Eur Econ Assoc* (2023).

[3] Vosoughi et al., "[The spread of true and false news online](#)", *Science* (2018).

[4] Lorenz-Spreen et al., "[A Systematic Review of Worldwide Causal and Correlational Evidence on Digital Media and Democracy](#)", *Nat Hum Behav* (2021).

[5] González-Bailón et al., "[Asymmetric ideological segregation in exposure to political news on Facebook](#)", *Science* (2023).

[6] Sabatini & Sarracino, "[Online Social Networks and Trust](#)", *Soc Indic Res* (2021).

[7] Lee, "[Connecting Social Media Use with Gaps in Knowledge and Participation in a Protest Context](#)", *Asian J Commun* (2019).

[8] Boy & Uitermark, "[On Display: Instagram, the Self, and the City](#)", *Ox Univ Press* (2023).

More Pragmatic Reasons

High enough agent density for emergent phenomena

- Traditionally high bot activity evidences existing incentives to deploy agents.
- Open protocols (ATproto, activity pub) reduce implementation friction.
- Lax moderation policies by major players increase probability of problematic outcomes.

There are some actual legal foundations to talk about

- AI Act
- Digital Services Act
- Beginning enforcement of both

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Commission fines X €120 million under the Digital Services Act

The Commission has issued a fine of €120 million to X for breaching its transparency obligations under the Digital Services Act (DSA)



The breaches include the deceptive design of its 'blue checkmark', the lack of transparency of its advertising repository, and the failure to provide access to public data for researchers.

Deceptive design of X's 'blue checkmark'

X's use of the 'blue checkmark' for 'verified accounts' deceives users. This violates the DSA obligation for online platforms to prohibit deceptive design practices on their services. On X, anyone can pay to obtain the 'verified' status without the company meaningfully verifying who is behind the account, making it difficult for users to judge the authenticity of accounts and content they engage with. This deception exposes users to scams, including impersonation frauds, as well as other forms of manipulation by malicious actors. While the DSA does not mandate user verification, it clearly prohibits online platforms from falsely claiming that



Hi, I am a physicist

- No legal background.
- But: Spent a lot of time thinking about the implications of the DSA for social media **recommendation algorithms**.
- Some insight into data access under DSA Art. 40(4) (tried it) and enforcement (hanging out with regulators).
- Have also thought a lot about studying AI swarms on social media recently.

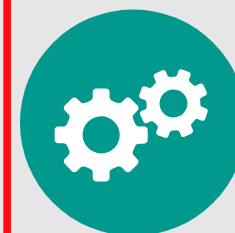
Designing social media recommendation algorithms for societal good (ERC StG)



Integrate people's judgements into algorithm development.



Balance freedom of expression and algorithmic interventions.



Develop Open Source digital twins to study algorithms.



Recommendations for platform regulation under the EU's DSA.

The AI Act

Providers and Deployers

*‘**provider**’ means a natural or legal person, public authority, agency or other body that develops an AI system or a general-purpose AI model or that has an AI system or a general-purpose AI model developed and places it on the market or puts the AI system into service under its own name or trademark, whether for payment or free of charge;*

Article 3(3), AI Act

*‘**deployer**’ means a natural or legal person, public authority, agency or other body using an AI system under its authority except where the AI system is used in the course of a **personal non-professional activity**;*

Article 3(4), AI Act



Transparency

*"Providers shall ensure that AI systems intended to interact directly with natural persons are **designed and developed** in such a way that the natural persons concerned **are informed that they are interacting with an AI system**, unless this is obvious from the point of view of a natural person [...]."*

Article 50(I), AI Act, in force since Aug 2026

*"AI agents are covered by Article 50(1) if they are **capable of interacting with the persons instructing them or with other natural persons in the execution of the tasks**. [...] This includes cases where AI agents are capable of interacting in complex **multi-agent architectures** with other AI agents that directly interact with the natural persons in the execution of the task."*

Examples: AI bots on social networks and media, coding agents.

Guidelines on transparency obligations for providers and deployers of AI systems

Systemic Risk

‘systemic risk’ means a risk that is specific to the high-impact capabilities of general-purpose AI models, having a significant impact on the Union market due to their reach, or due to actual or reasonably foreseeable negative effects on public health, safety, public security, fundamental rights, or the society as a whole, that can be propagated at scale across the value chain;

Article 3(65), AI Act

Systemic risks include any actual or reasonably foreseeable negative effects in relation to major accidents, disruptions of critical sectors and serious consequences to public health and safety; any actual or reasonably foreseeable negative effects on democratic processes, public and economic security; the dissemination of illegal, false, or discriminatory content.

Recital 110, AI Act



Mitigating Systemic Risk

[...] providers of general-purpose AI models with systemic risk shall:

- (a) perform **model evaluation** [...] including [...] **adversarial testing** of the model [...];*
- (b) **assess and mitigate possible systemic risks** at Union level [...];*
- (c) keep track of, document, and report, without undue delay [...] relevant information about **serious incidents** and possible corrective measures to address them;*
- (d) ensure an **adequate level of cybersecurity** protection for the general-purpose AI model with systemic risk and the physical infrastructure of the model.*

Article 55(1), AI Act, in force since Aug 2025

Prohibited AI Practices

"The following AI practices shall be prohibited:

*the placing on the market, the putting into service or the use of an AI system that deploys subliminal techniques beyond a person's consciousness or purposefully manipulative or deceptive techniques, with the objective, or **the effect of materially distorting** the behaviour of a person or a group of persons by appreciably impairing their ability to make an informed decision, thereby causing them to take a decision that they would not have otherwise taken in a manner that causes or is reasonably likely to cause that person, another person or group of persons **significant harm**;"*

Article 5, AI Act, in force since Feb 2025

Harmful Manipulation as Systemic Risk

*Harmful manipulation: Risks from enabling the **strategic distortion of human behaviour or beliefs** by targeting large populations or high-stakes decision-makers through persuasion, deception, or personalised targeting. This includes significantly enhancing capabilities for persuasion, deception, and personalised targeting, particularly through multi-turn interactions and where individuals are unaware of or cannot reasonably detect such influence. Such capabilities could undermine democratic processes and fundamental rights, including exploitation based on protected characteristics.*

The General-Purpose AI **Code of Practice**

Regulatory Gaps in the AI act

- **Emergence**: Risk arises from interaction, not from intended purpose.
- **Collective harm**: The prohibitions and fundamental-rights framing are individual-centric.
- **Agent-to-agent scenarios** are mentioned nowhere explicitly.
- **Personal-use** exempts scenarios with delegates that, in aggregate, could still lead to emergent harmful behaviour.
- **Significant harm** is a high barrier to cross for deceptive and manipulative techniques to be prohibited. The code of practice is not very binding.
- **Enforcement against bad actors** (deployers) is unrealistic. Realistic targets are providers and downstream systems (see DSA).

The Digital Services Act

Very Large Online Platforms and Search Engines (VLOPs/VLOSEs)

Online platforms and online search engines which have a number of average monthly active recipients of the service in the Union equal to or higher than 45 million.

Currently: X, TikTok, Facebook, Instagram, YouTube, Snapchat, LinkedIn, Pinterest, Reddit, Whatsapp, **ChatGPT**.

Identifying systemic risks

Providers [...] shall diligently identify, analyse and assess any systemic risks [...] stemming from the design or functioning of their service [...] including algorithmic systems [...].

The risk assessment [...] shall include the following systemic risks:

- dissemination of illegal content
- negative effects on fundamental rights
 - human dignity
 - protection of personal data
 - freedom of expression and information
 - non-discrimination
 - consumer protection
- negative effects on **civic discourse, electoral processes**, public security
- gender-based violence, physical and mental wellbeing
- public health and minors

Intentional Manipulation

The assessments shall also analyse whether and how the risks [...] are influenced by intentional manipulation of their service, including by inauthentic use or automated exploitation of the service, as well as the amplification and potentially rapid and wide dissemination of illegal content and of information that is incompatible with their terms and conditions.

Mitigating systemic risks

Providers [...] shall put in place reasonable, proportionate and effective mitigation measures [...]. Such measures may include [...]: testing and adapting their algorithmic systems, **including their recommender systems** [...]; adapting their terms and conditions and their **enforcement**.

Synthesis

- Regulation via systemic risks is **more durable given rapidly changing technologies** but also subject to **translation into concrete designs and metrics** to become enforceable.
- Systemic risk definitions give **normative grounding** for research endeavours.
- **Systemic risk definitions** are much broader in the DSA (civic discourse, intentional manipulation) than in the AI act.
- Social media platforms rely on the **observability** of agentic activity to act.

Observability of Agentic Activity

- Article 40(12) DSA allows researchers to request **public data** from VLOPs/VLOSEs for the purpose of researching systemic risks under the DSA.
- Article 40(4) DSA allows researchers to request **private data** for the same purpose.
- Article 40(12) requests are starting to work. Article 40(4) not so much.

Could we use this to study the patterns of agentic activity and develop warning signs?

How can we drive change?

- Provide the regulator with **evidence to enforce** the DSA and AI Act.
- Develop definitions and metrics to **inform a delegated act on multi-agent systemic risk** for the AI Act.
- Catalogue **best practices** that providers of general purpose AI can adopt to show compliance.
- Develop **European model alternatives** and wait for a competitive advantage through enforcement.

Open questions

- Which concrete design decisions does the mitigation of systemic risks in the DSA and AI Act map to?
- Which signals are most promising to inform early warning signs of AI swarm activity on social media?
- Are the risks posed by AI swarms on social media platforms just "more of the same" or of a new quality?
- Is there any hope to label AI generated content at scale with enough accuracy?
- What are other real-world systems where we expect substantial AI swarms to show up?